

L5 - Insertion Sort

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Conceptual Framework:

1. Divide the list into two parts, sorted part & unsorted part
2. Insert one element from the unsorted part into the sorted part
 - a. Find the correct position in the sorted part to insert the element
 - b. Make room for the new element
 - c. Place the new element in the correct position
3. Repeat step two to until the entire list is sorted

5 2 4 6 3 1

6 5 4 3 2 1

	k	j						
1	2	3	4	5	6	sorted list	1..j - 1	
5	2	4	6	1	3	key = 2		

	k	j						
1	2	3	4	5	6	sorted list	1..j - 1	
2	5	4	6	1	3	key = 4		

	k	j						
1	2	3	4	5	6	sorted list	1..j - 1	
2	4	5	6	1	3	key = 6		

	k	j						
1	2	3	4	5	6	sorted list	1..j - 1	
2	4	5	6	1	3	key = 1		

	k	j						
1	2	3	4	5	6	key = 1		
2	4	5	6	6	3			

	k	j						
1	2	3	4	5	6	key = 1		
2	4	5	5	6	3			

InsertionSort(Arr)

1. for j=2 to Arr.length
2. let key = Arr[j]
3. let k = j - 1
4. while k > 0 and Arr[k] > key
5. Arr[k+1] = Arr[k]
6. k = k - 1
7. Arr[k+1] = key

k j
 1 2 3 4 5 6 key = 1
 2 4 4 5 6 3

k j
 1 2 3 4 5 6 key = 1
 2 2 4 5 6 3

k j
 1 2 3 4 5 6 key = 1
 1 2 4 5 6 3 Arr[k+1]=key

k j
 1 2 3 4 5 6 sorted list 1 .. j - 1
 1 2 4 5 6 3 key = 3

k j
 1 2 3 4 5 6 key = 3
 1 2 3 4 5 6 Arr[k+1] = key

j
 1 2 3 4 5 6 sorted list 1 .. j - 1
 1 2 3 4 5 6

InsertionSort(Arr)	cost	freq
1. For j=2 to Arr.length	c1	n
2. let key = Arr[j]	c2	n-1
3. let k = j - 1	c3	n-1
4. while k > 0 and Arr[k] > key	c4	$(n+2)(n-1)/2$ (n-1)
5. Arr[k+1] = Arr[k]	c5	$n(n-1)/2$ (0)
6. k = k - 1	c6	$n(n-1)/2$ (0)
7. Arr[k+1] = key	c7	n-1

Worst case Analysis:

Worst Case occurs when the input list is already sorted in the reverse order of the desired order.

6 5 4 3 2 1 (List already sorted in reverse order)

5 6 4 3 2 1 (1+1)

4 5 6 3 2 1 (2+1)

3 4 5 6 2 1 (3+1)

2 3 4 5 6 1 (4+1)

1 2 3 4 5 6 (5+1)

Taking Sum

$2+3+4+5+\dots+(n-1)+n$

$(n+2)(n-1)/2$

For lines 5 & 6, the series is $1+2+3+4+5+\dots+n-1=n(n-1)/2$

$T(n) = c_1(n) + (c_2+c_3+c_7)(n-1)/2 + (c_5+c_6)(n(n-1)/2) + c_4(n+2)$
 $(n-1)/2$

$T(n) = c_{10}(n^2) + c_{11}n + c_{12}$

$T_w(n) = O(n^2)$

Best case:

Best Case occurs when the list is already sorted in the desired/required order.

1 2 3 4 5 6 (List already sorted in required order.)

1 2 3 4 5 6 (1)

1 2 3 4 5 6 (1)

1 2 3 4 5 6 (1)

1 2 3 4 5 6 (1)

1 2 3 4 5 6 (1)

$T(n) = c_1(n) + (c_2+c_3+c_4+c_7)(n-1)$

$T(n) = c_{11}n + c_{12}$

$T_b(n) = \Omega(n)$

$$T(n) = \theta(n^2)$$

$$T(n) = \theta(n) \quad \mathbf{x}$$